

Heath Street East Pedestrian Bridge

Proposition to Rebuild

1.0 Introduction

The neighbourhood surrounding the Moore Park Ravine is comprised of many Victorian style homes, with a quiet atmosphere. Spanning the ravine is the Heath St. East pedestrian bridge, known fondly by locals as the Cat's Eye bridge. As this bridge is used frequently by the residents of the area, our team believes this would be a good opportunity for revitalizing local infrastructure. This report is a summary of the considerations taken when designing the replacement bridge. The following brief discusses the affected stakeholders, as well as requirements, objectives, metrics, constraints, and criteria used by the design team. Furthermore, key design features will be summarized and justified.

2.0 Stakeholders

2.1. Primary Stakeholder

The primary stakeholders are the residents of the area as well as those who frequent the current bridge. These stakeholders will be the main focus of this proposition as they are directly impacted by the construction and quality of the bridge in their daily lives. This proposal aims to improve the lives of the locals by refining the quality of local infrastructure and enhancing resident connection with nature.

2.2. Secondary Stakeholder

The secondary stakeholder in this situation is the City of Toronto as they will be funding the project. In order to satisfy the desires of the City, this bridge design was created to be cost effective, while improving the municipal infrastructure and recreational value.

3.0. Requirements

The use of the words “must”, “must not”, “should”, and “should not” in this and following sections are to be defined as in RFC 2119 [1].

3.1. Objectives

The high level objective of this report is to provide a safe and cost effective means of improving the quality of infrastructure, and enhancing the recreational value of the area. Detailed objectives to ensure that the high-level objective have been met include: minimizing material cost, improving functionality, and providing an aesthetically pleasing connection with the surrounding ravine.

3.2. Metrics

The quality of the infrastructure in this design is measured by the dimensions of the passageway, as the functionality of the bridge is dependent on its ability to accommodate both pedestrians and cyclists. The metric for affordability is the cost of materials as this will be the

main expenditure in the design of the new bridge. Finally, the metric for aesthetics is the length of the bridge which provides an unobstructed view of the ravine and tree canopy. This ensures that users are able to experience the full scope of Toronto's natural beauty.

3.3. Constraints

The length of the bridge must be no less than 80 m to span the width of the ravine. The width of the pedestrian walkway must be no less than 3 m to comply with City of Toronto building codes [2]. The bridge must be able to withstand a live load of 5 kN/m, a wind force of 2 kN/m², as well as loading due to its own self weight. No section of the bridge must have a mid-span deflection caused by live load greater than the length of the section divided by 300.

3.4. Criteria

While designing this bridge, several criteria were considered. Pertaining to cost, the design should minimize the expenditure on materials. As well, the new bridge should work to maximize pathway width to accommodate pedestrians and cyclists to pass comfortably. Finally, to improve aesthetics, a longer section of the bridge that is open to nature was preferred.

4.0. Key Design Features

Our proposed bridge design aims to feature the beautiful surroundings of the bridge. Unfortunately, the truss design of the current bridge distracts from the trees in the area. In order to allow pedestrians to admire the surrounding nature, an inverted truss design was chosen to minimize view obstruction. The middle of the bridge was chosen to be an inverted truss, while the ends of the bridge are non-inverted to provide dimensional contrast.

The width of the proposed plan is an increase of more than 150% of the current bridge. This works to accommodate both cyclists and pedestrians, while fitting within the dimensions of the existing path.

A Pratt design was chosen for the trusses as it is symmetrical and aesthetically pleasing. This was chosen over a K or bowstring truss in order to minimize the obstructions caused by the structure.

As well, the lighting in the current bridge is very poor, with only two dim lamps, one on each support pillar. To fix this, the proposed design includes 60 lights to be installed in the deck along both sides of the span. The chosen lights are the LineLight R2 by SolarPath as they are durable and are able to change colours for special occasions [3].

A childproof steel railing from Weland products has been chosen as it is aesthetically pleasing, safe for pedestrians, and cost-effective [4]. This will improve functionality of the bridge, as well as decrease the accessory cost.

5.0. Conclusion

Therefore, our bridge design will accentuate the natural beauty of the neighbourhood as well as improve the infrastructure of the area. Our design should be chosen amongst others, as it will revitalize the area with respect to nature, recreation, and structural significance.

Source Extracts

[1] S. Bradner. *Key Words for Use in RFC's to Indicate Requirement Levels*. [Online] Available: <https://www.ietf.org/rfc/rfc2119.txt>, 1997.

Network Working Group
Request for Comments: 2119
BCP: 14
Category: Best Current Practice

S. Bradner
Harvard University
March 1997

Key words for use in RFCs to Indicate Requirement Levels

Status of this Memo

This document specifies an Internet Best Current Practices for the Internet Community, and requests discussion and suggestions for improvements. Distribution of this memo is unlimited.

Abstract

In many standards track documents several words are used to signify the requirements in the specification. These words are often capitalized. This document defines these words as they should be interpreted in IETF documents. Authors who follow these guidelines should incorporate this phrase near the beginning of their document:

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in RFC 2119.

Note that the force of these words is modified by the requirement level of the document in which they are used.

1. **MUST** This word, or the terms "REQUIRED" or "SHALL", mean that the definition is an absolute requirement of the specification.
2. **MUST NOT** This phrase, or the phrase "SHALL NOT", mean that the definition is an absolute prohibition of the specification.
3. **SHOULD** This word, or the adjective "RECOMMENDED", mean that there may exist valid reasons in particular circumstances to ignore a particular item, but the full implications must be understood and carefully weighed before choosing a different course.
4. **SHOULD NOT** This phrase, or the phrase "NOT RECOMMENDED" mean that there may exist valid reasons in particular circumstances when the particular behavior is acceptable or even useful, but the full implications should be understood and the case carefully weighed before implementing any behavior described with this label.

[2] City of Toronto. *Accessibility Design Guidelines*. [Online] Available: https://www1.toronto.ca/static_files/equity_diversity_and_human_rights_office/pdf/accessibility_design_guidelines.pdf, 2004.

All pedestrian routes should be safe and easy to use by a wide range of persons with disabilities. Generally, such routes should be easily identifiable, clearly separated from vehicular routes, and free of obstacles at all times of the year.

1.1.1 Bridges with Pedestrian Access

- Pedestrian sidewalks located on either side of bridges, should have slopes no greater than 1:20 (5%), and cross slopes no greater than 1:50, or (2%), wherever possible.
- No sidewalk on a bridge should be less than 1220 mm wide where minimal pedestrian traffic is expected, and a minimum of 1525 mm wide where frequent pedestrian is anticipated (e.g., two-way pedestrian traffic).
- Handrails or guards located beside sidewalks should be easy to grasp at 50 mm wide (maximum) and mounted at a suitable height. (See Figure 1)
- It is recommended that pedestrian footbridges have ramp access rather than stairways.

1.1.2 Crosswalks

use one crosswalk to cross at right angles to one another. All such pedestrian routes should be free of obstacles, such as light standards, traffic signal supports, posts or catch basins as well as

provide access to storage containers. (See Section 1.5.6 Traffic Signals and Figure 69)

Figure 1

[3] SolarPath. *Linelight R2*. [Online] Available: <http://www.solarpathusa.com/linelight-r2/>, 2015.

SolarPath
sun solutions

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TECHNOLOGY AND OPERATION

Energy Source	– Solar modules – External DC feed – Hybrid – Solar & External DC backup feed
Energy consumption	12V-24V DC, 160mA
Illumination technology	12 x Ultra-bright 73mm LEDs.
Operating temperature range	-22F to +185F / -30C to +85C
Warranty	2 years

ILLUMINATION PROPERTIES

Available Flash Patterns	Controllable – Flashing, Steady or Sequential
Available colors Lighting intensity (mcd)	• White: 6400 • Blue: 3200

[4] Weland. *Universal Steel Railing*. [Online] Available: http://www.weland.se/en-gb/raecke_universal_stal/, 2017.

Childproof railing

Hot dip galvanised.

Balusters/hand railing Ø 42 mm

Bottom rails: Ø 27 mm

Intermediate balusters: Ø 12 mm, max opening between balusters 100 mm

Height: 1100 mm

Length (mm)	Height (mm)	Stocked	Part No.	Weight (kg/each)
365	1100	X	UB0442	8,9
465	1100	X	UB0542	10,4
665	1100	X	UB0742	13,3
965	1100	X	UB1042	17,8
1465	1100	X	UB1542	25,0
1965	1100	X	UB2042	32,1

These products are available:

Stocked

Source Extracts

- [1] V. Pahwa. *The Heath Street East Pedestrian Bridge Over Mud Creek*. [Online] Available: <http://vikpahwa.com/2012/09/>, 2012.



- [2] J. Larson. *How Does Nature Impact our Wellbeing*. [Online] Available: <https://www.takingcharge.csh.umn.edu/enhance-your-wellbeing/environment/nature-and-us/how-does-nature-impact-our-wellbeing>, 2016.



Nature heals

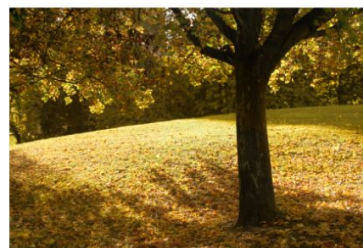
Being in nature, or even viewing scenes of nature, reduces anger, fear, and stress and increases pleasant feelings. Exposure to nature not only makes you feel better emotionally, it contributes to your physical wellbeing, reducing blood pressure, heart rate, muscle tension, and the production of stress hormones. It may even reduce mortality, according to scientists such as public health researchers Stamatakis and Mitchell.

Research done in hospitals, offices, and schools has found that even a simple plant in a room can have a significant impact on stress and anxiety.

Nature soothes

In addition, nature helps us cope with pain. Because we are genetically programmed to find trees, plants, water, and other nature elements engrossing, we are absorbed by nature scenes and distracted from our pain and discomfort.

This is nicely demonstrated in a now classic study of patients who underwent gallbladder surgery; half had a view of trees and half had a view of a wall. According to the physician who conducted the study, Robert Ulrich, the patients with the view of trees tolerated



[3] Google Maps. [Online] Available:
https://www.google.ca/maps/place/Cat's+Eye+Bridge/@43.6936729,-79.3780414,3a,75y,142.53h,102.81t/data=!3m6!1e1!3m4!1sobqnJ53RRS-M_fffbYtmVA!2e0!7i13312!8i6656!4m5!3m4!1s0x89d4ccb4fea8313f:0x52bb401d87807aa!8m2!3d43.693554!4d-79.3778775, 2017.

